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FitTrack AI

Personalized Fitness Recommendations Powered by AI

PHASE 7

PROJECT DOCUMENTATION

Project Name: AI FitTrack

Project Type: AI-powered Fitness Tracking Backend / RESTful API Application

Technology Stack: Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt.js, Google Gemini AI

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7.1 Documentation Overview

The documentation phase consolidates the technical and functional information relating to FitTrack AI – Personalized Fitness Recommendations Powered by AI into a structured project report. It records the application's purpose, requirements, system design, database structure, implementation modules, API functionality, testing process, and project outcomes.

The report is intended to provide readers with a clear account of how the AI FitTrack backend was designed, developed, and tested. It also serves as a reference for understanding the system's components and their interactions.

7.2 Project Introduction and Background

The introductory portion of the project report presents the project title, background, problem identification, proposed solution, and objectives. Together, these sections establish the purpose of AI FitTrack and explain the need for a centralized fitness tracking system supported by AI-assisted recommendations.

The project addresses the difficulty of maintaining workout records manually. It provides backend functionality for managing workout information and delivering personalized fitness guidance through an integrated AI service.

7.3 Requirement Documentation

This section records the software and hardware requirements identified for project development and testing. The software environment includes the runtime, backend framework, database, development environment, and API testing tools.

7.3.1 Software Requirements

Software / Tool	Purpose
Node.js	JavaScript runtime for the backend application.
npm	Package management and dependency installation.
Express.js	Framework for implementing RESTful API routes and server functionality.
MongoDB	Database for storing user and workout information.
Mongoose	Object data modeling and interaction with MongoDB.
Visual Studio Code	Development environment for writing and managing project code.
Postman or Thunder Client	API request execution and response verification.

The project also uses JWT for authentication, bcrypt.js for password hashing, and Google Gemini AI for AI-assisted fitness functionality.

7.3.2 Hardware Requirements

The hardware requirements cover the processor, RAM, and storage needed for development and testing. Exact minimum or recommended specifications should be entered according to the actual development environment; they are not specified in the supplied phase content.

Hardware component	Specification
Processor	[ENTER PROCESSOR SPECIFICATION]
RAM	[ENTER RAM CAPACITY]
Storage	[ENTER AVAILABLE STORAGE]

7.4 System Design Documentation

The system design documentation describes the layered MVC-based architecture used by AI FitTrack. The principal components are the client layer, Express server, API routes, authentication middleware, controllers, services, database models, and Gemini AI service.

A client request is received by the Express server and routed to the relevant API endpoint. Where required, authentication middleware validates the request's authentication context before the controller coordinates application logic. Services handle supporting operations, database models provide access to MongoDB data, and the Gemini AI service supports AI-powered functionality. The backend returns the result to the client as a JSON response.

7.5 Database Documentation

The database documentation describes the MongoDB collections, their fields, validation requirements, and associations. The two primary collections identified for the project are Users and Workouts.

7.5.1 Users Collection

The Users collection stores user-related information required by the application, including information used for authentication and profile functionality. The exact schema fields and validation rules should be

7.5.2 Workouts Collection

The Workouts collection stores workout records managed through the workout APIs. The implemented Mongoose workout model should be used as the authoritative source for listing field names, data types, and validation requirements.

7.5.3 Relationship Between Collections

A one-to-many relationship exists between Users and Workouts: one user can have multiple workout records. The database documentation should identify how each workout references its associated user, based on the implemented schema.

7.6 Implementation Documentation

The implementation section describes the major backend modules and their responsibilities. The following modules are identified in the project:

Module	Documented responsibility
Server configuration	Initializes and configures the backend server.
Database configuration	Establishes the application's connection to MongoDB.
User authentication	Provides registration, login, and profile-related functionality.
Workout management	Supports workout record creation, retrieval, updating, deletion, and searching.
AI recommendation service	Integrates AI-assisted personalized workout recommendations.
Authentication middleware	Protects endpoints that require an authenticated user.
Error handling	Supports the handling of errors arising during API processing.

The description of each module should be aligned with the actual source code and configuration used in the project.

7.7 API Documentation

The API documentation provides an organized reference to the services exposed by the backend. Endpoint paths, HTTP methods, request formats, authentication requirements, and response examples should be recorded from the implemented routes and verified API requests.

7.7.1 Authentication APIs

Authentication APIs provide user registration, login, and profile-related functionality. The profile endpoint is protected and is accessed using the authentication token obtained after login.

7.7.2 Workout APIs

Workout APIs support the following operations:

- Add workout records.
- Retrieve all workouts for the authenticated user.

- Retrieve a workout by its identifier.
- Update an existing workout record.
- Delete a workout record.
- Search workout records.

7.7.3 AI APIs

AI APIs provide personalized workout recommendations and fitness insights through the integrated Google Gemini AI service. The documentation should include the actual input requirements and representative outputs obtained from testing.

7.8 Testing Documentation

The testing section records API verification performed using an API testing tool such as Postman or Thunder Client. Testing is organized into three functional groups:

1. Authentication API testing
2. Workout API testing
3. AI API testing

For each test, the report should record the request, relevant input data, response, and observed result. Test outcomes should reflect actual execution and should not be inferred from the planned test description.

7.9 Supporting Evidence

Relevant project evidence may be incorporated into the appropriate report sections. Evidence should be placed near the feature or process it supports, with a figure number and descriptive caption.

- Implementation and project-structure screenshots.
- API testing requests, responses, and results.
- Database-related screenshots.
- AI-generated recommendation and fitness-insight outputs.

Only genuine screenshots and outputs captured from the project should be inserted. Each item should be legible and directly related to the accompanying explanation.

7.10 Final Documentation Structure

The complete project documentation presents the project phases in chronological order, from initial ideation through the final demonstration.

1. Brainstorming & Ideation

2. Requirement Analysis

3. Project Design

4. Project Planning
5. Project Development
6. Project Testing
7. Project Documentation
8. Project Demonstration

This sequence provides a continuous view of the project lifecycle, covering the initial concept, requirements, design, planning, implementation, testing, documentation, and demonstration.

7.11 Phase Summary

This phase establishes the structure and content of the AI FitTrack project report. It brings together the project background, requirements, system architecture, database design, implementation modules, API functionality, testing approach, and supporting evidence. Specific schema details, hardware specifications, endpoint definitions, screenshots, and test results should be completed using the actual project artifacts to maintain accuracy.